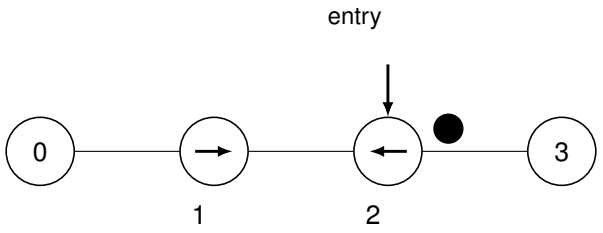
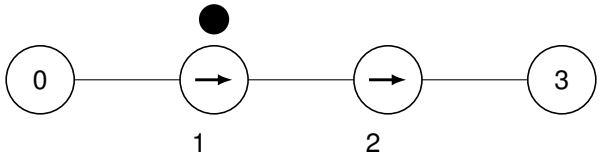
On a road, a counter may revisit a circle. Follow its NEXT arrow, then flip that card, every time. Stop at either end. Finish one counter before launching the next; keep the cards between counters. Keep the counter beside the card so its arrow stays visible. A partner checks each move.

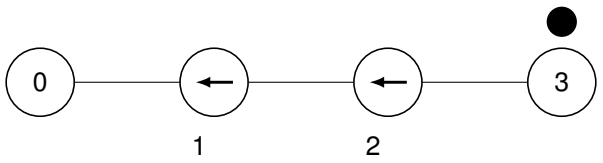
Example: enter at 2 on this short road.



Input: counter at 2.
Cards at 1 and 2: R, L.



After $2 \to 1$:
counter at 1; cards R, R.

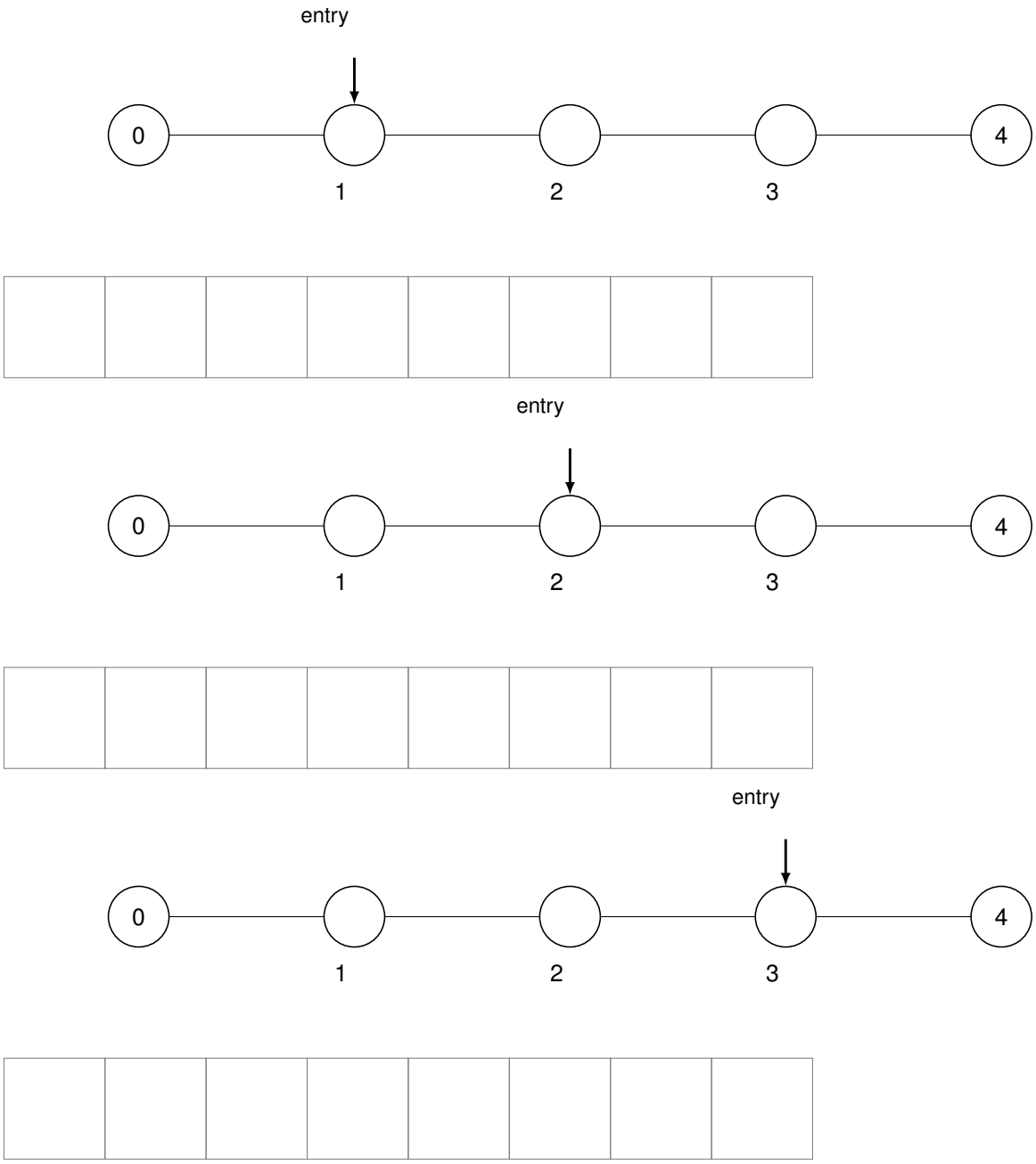


Then $1 \to 2 \to 3$.
Exit: 3. Cards: L, L.

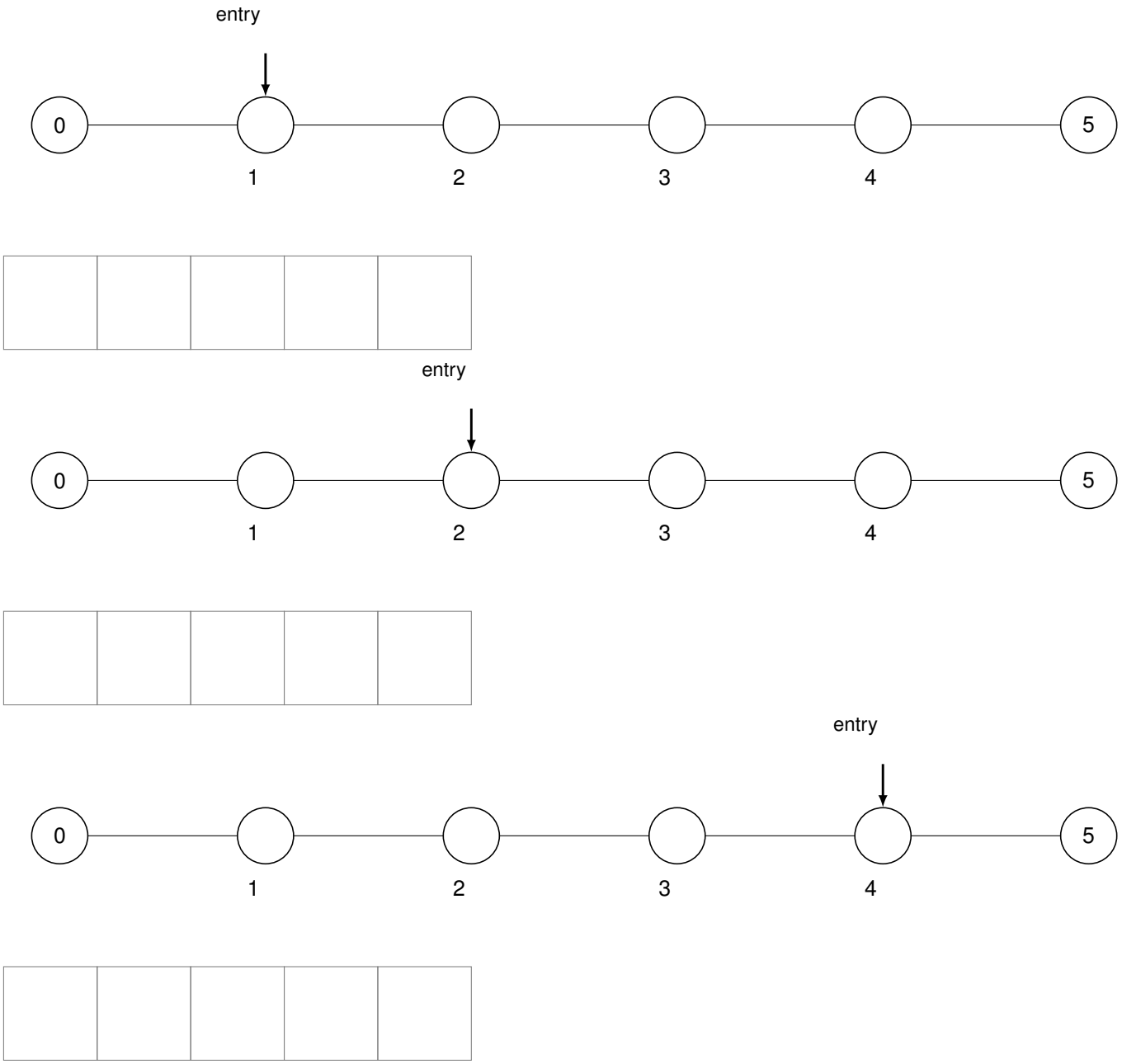
Problem 1: Use the road from 0 to 4. Enter at 1, with starting cards L, L, R at 1, 2, 3. Send eight counters. Keep the arrival order. Where will the next four counters exit?

Arrival order

Problem 2: On the road from 0 to 4, find out how moving the entry changes the exits. Choose any starting arrows for each new trial. Send eight counters from each marked entry.



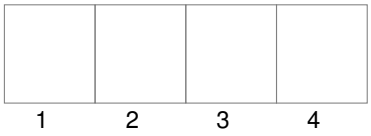
Problem 3: Use the road from 0 to 5. Can you choose starting arrows so that all five counters exit at 5? Try entry 1, entry 2 and entry 4. Find a rule for the number reaching each end in five launches.



Problem 4: Choose a road and an entry. Can the exit counts repeat in equal blocks while the complete arrow positions change? Find an example or explain why none can exist.

Road: _____ Entry: _____ Block size: _____

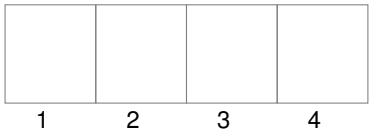
Starting arrows



After one block



After two blocks



Position 4 is only for the road from 0 to 5.

Arrival order



Problem 5: Imagine a longer road: $0, 1, 2, \ldots, m$. Always enter at the same inside circle, s. Predict how many of m counters reach each end, for any starting arrows. Explain why your rule works.

Problem 6: Could a counter travel forever without reaching either end? Explain your answer for any finite road.